Context Without Contact: Top-Down Information Flow from Shared Modulation in Uncoupled Prebiotic Systems

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- **0Abstract**

Shifts from bottom-up to top-down influences are considered a hallmark of major evolutionary transitions, including the origin of life. We explore how such directional information flow can arise without direct interactions among the system components using a minimal model of uncoupled logistic maps subject to a shared, time-varying constraint. Transfer entropy (TE) reveals a consistent macro-to-micro bias across diverse modulation profiles, along with increased inter-unit predictability and confinement of macroscopic dynamics to low-dimensional manifolds. Conditional transfer entropy (CTE), conditioning on the environmental signal, reveals that a macro-to-micro asymmetry remains even after the influence of the shared driver is accounted for. This persistence suggests that nonlinear aggregation of uncoupled chaotic units generates predictive structure beyond the common environmental forcing. These results show that structured environmental variation alone is sufficient to generate an asymmetric top-down directional information flow, suggesting that dynamic external constraints may serve as an early scaffold for coordination before the evolution of internal coupling mechanisms.

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Introduction

Top-down causation, in which higher-level organization shapes the dynamics of lower-level components, occurs at all scales of biological organization and helps explain the hierarchy of structures in living systems (1-3). It has been proposed that shifts from predominantly bottom-up to predominantly top-down influences are hallmarks of major evolutionary transitions (4,5), including the emergence of life itself (6-8). A taxonomy of top-down causation mechanisms has been proposed, distinguishing algorithmic, non-adaptive, and adaptive forms, with the case made that downward causation is essential for understanding biological organization (6). The major evolutionary transitions have been identified as events in which formerly independent replicating entities become integrated into higher-level units, implying the emergence of top-down constraints (4), a framework further extended to include the informational and regulatory dimensions of these transitions (5). Top-down causation by information control has been framed as a scientific research programme providing a philosophical foundation for empirical investigation (2), and the origin of life has been characterized as a transition from dynamics governed solely by physical law to dynamics governed by information (7,8). One way to study such transitions is by examining directional information flow between scales, specifically whether the macroscopic state of a system can predict the dynamics of its parts better than the reverse (9).

This perspective motivates a key question particularly relevant for early chemical evolution: can asymmetric directional information flow between levels arise even in the absence of direct coupling among low-level components, driven solely by a shared environmental structure? Structured external cycles (such as tides, day-night rhythms, or wet-dry fluctuations) are known to promote essential prebiotic processes (10-12), but could such global patterns have also imposed system-wide constraints capable of coordinating dynamics across otherwise independent units?

To investigate this question, we employ logistic maps as our model system. The logistic map, originally introduced as a minimal model of population dynamics (13), has become a paradigmatic tool in nonlinear dynamics and computational biology. Despite its simplicity, it captures the essential features of density-dependent growth, including period-doubling cascades and deterministic chaos (14). More broadly, the synchronization of chaotic systems under common external forcing is a well-characterized phenomenon in nonlinear dynamics (15); even without mutual coupling, a shared driving signal can entrain individually chaotic units into coherent collective behavior. Coupled logistic maps have been widely used to study synchronization, collective behavior, and information flow in spatially extended systems (16,17). In the context of prebiotic chemistry, such minimal (albeit abstract) models can be extremely valuable because they isolate the dynamical consequences of environmental forcing from the biochemical complexity of real systems, allowing the identification of generic mechanisms that do not depend on molecular detail (18).

We use a deliberately minimal model in which a population of uncoupled logistic maps is subject to a single time-varying global constraint, implemented as a sinusoidal modulation of the carrying capacity. We then investigated whether such purely environmental structuring can produce measurable macro-to-micro predictive bias. We computed Transfer Entropy (TE) (19) to quantify directional information flow. TE measures how much knowledge of the past of one variable improves predictions about the future of another, beyond what the variable's own past can explain. TE has become a standard tool for detecting directed information flow in complex systems and has been applied extensively in neuroscience to map effective connectivity between brain regions (20,21), in climate science to identify causal interactions between climatic variables (22,23), and in ecology to characterize predator-prey and species interaction networks (24). In the context of multi-scale biological systems, TE has been used to quantify information flow between levels of organization, including the detection of top-down informational influence in coupled map lattices (9,16). A key limitation of standard TE is its sensitivity to common-driver confounds: when two variables share a common input, TE may detect apparent information transfer that is entirely attributable to the shared source (20,22,25). Conditional transfer entropy (CTE), which conditions out

known confounding variables, provides a principled correction for this issue (23,26). We employ both unconditional and conditional TE in our analysis to distinguish directional predictability arising from shared environmental forcing from residual population-level structure.

While the interpretation of informational asymmetry as literal causation remains debated, here we treat it as an operational signal or a measurable statistical signature that may accompany early forms of coordinated behavior, regardless of whether an underlying causal mechanism is in place. This framework has been used to operationalize top-down informational influence in a system where units are explicitly coupled (9); however, here, the focus is on the limiting case with no internal coupling, isolating environmental structure as a potential driver of coordination in prebiotic contexts.

Model

We consider a minimal system of *N* uncoupled logistic maps, each evolving under a shared time-dependent carrying capacity, K_n. Let $x_{i,n}$ denote the state of unit *i*, where $i = \{1 \ldots N\}$, at discrete time step *n*. Each unit follows:

$$x_{i,n+1} = r_i x_{i,n}\left(1 - \frac{x_{i,n}}{K_n}\right),$$

where $r_i \in [r_{min}, r_{max}]$ is sampled independently from a uniform distribution to ensure that, in the absence of modulation, each unit operates in a chaotic regime (9,16). There are no interactions among the units; the only shared influence is the global carrying capacity K_n, which is modulated sinusoidally as $K_n = K_0 + A\, sin(\omega\, n)$. This modulation represents synchronized environmental rhythms (e.g., day-night cycles, tides, or wet-dry transitions) that affect the entire population. The setup was inspired by an earlier toy model in which top-down control emerged from explicit mean-field coupling (9). In the present case, all units are independent and subject to a common time-varying environment. The macroscopic variable is defined as the mean field: $\mathrm{M}_n = \frac{1}{N}\sum_{i=1}^{N} x_{i,n}$.

While abstract, the model components can be loosely interpreted in prebiotic terms: the global constraint K_n represents an externally imposed limit on the system carrying capacity (e.g., availability of resources, space, or favorable conditions), the intrinsic parameters r_i reflect heterogeneity in how individual entities respond to that environment, and the mean field M_n captures an aggregate property of the population, such as an overall reaction yield or bulk chemical state.

Results

Unconditional transfer entropy reveals directional asymmetry in information flow

The time-varying K_n imposes a common oscillatory envelope on all trajectories x, introducing a coherent pattern that shapes the mean field M (Figure 1A). To assess whether this shared modulation produces directional relationships between macro- and micro-level dynamics, we computed the transfer entropy (TE) in both directions: top-down (from mean field to representative unit, $M \rightarrow x$) and bottom-up (from representative unit to mean field, $x \rightarrow M$). Formally:

$$T_{Y \rightarrow X} = \sum p\left(x_{n+1}, x_n^{(k)}, y_n^{(k)}\right) log_2 \frac{p\left(x_{n+1} | x_n^{(k)}, y_n^{(k)}\right)}{p\left(x_{n+1} | x_n^{(k)}\right)}$$

where $x_n^{(k)}$ and $y_n^{(k)}$ are the histories of length *k* for the target and source variables, respectively.

TE analysis revealed clear directional asymmetry (Figure 1B). Without modulation ($A \approx 0$), TE is negligible in both directions, which is consistent with unstructured dynamics. As *A* increases, the TE from the mean field to the individuals ($M \rightarrow x$) increases steadily. This indicates that shared modulation generates top-down informational influences without reciprocal predictability. The monotonic increase of TE($M \rightarrow x$) arises because larger modulation amplitudes strengthen the shared environmental component relative to unit-specific chaotic fluctuations, causing individual trajectories to become progressively more aligned with the mean field. In contrast, the reverse TE ($x \rightarrow M$) remains small and tends to decrease with

increasing modulation amplitude, reflecting the fact that the population mean increasingly averages out unit-specific fluctuations.

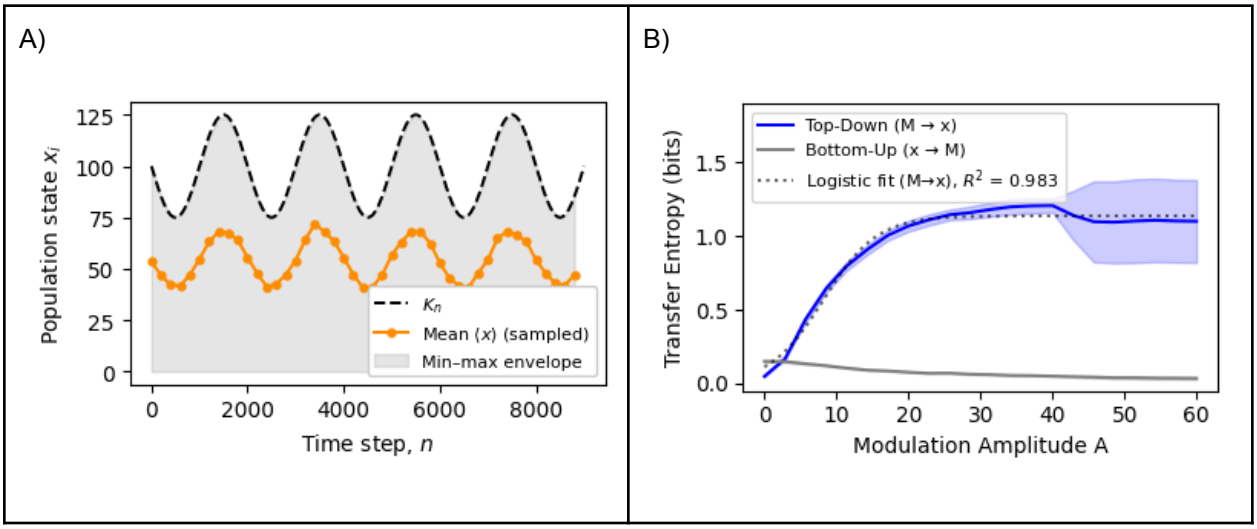

Figure 1. Modulation-driven asymmetry and entrainment under sinusoidal forcing.

(A) Time series of a population of *N* = 1000 uncoupled logistic units evolving under sinusoidal carrying capacity ($K_0 = 100$, $A = 25$, $r_i \in [3.9, 4.0]$). The black, dashed curve shows the external modulation K_n, the orange line the mean field *M* sampled every 200 steps, and the shaded band the range of individual states at each sample. (B) Transfer entropy (TE) between the mean field M_n and a representative unit $x_{i,n}$, computed in both directions: $M \to x$ (blue) and $x \to M$ (gray), across modulation amplitudes *A*. The dashed line shows the logistic fit to TE($M \to x$). Results are averaged over 10 seeds (*N* = 1,000, $K_0 = 100$, $r_i \in [3.9, 4.0]$, and *T* = 10,000 steps, with the first 1,000 discarded). Shaded regions show 95% confidence intervals.

At high modulation amplitudes, TE (M→x) approaches a plateau (Figure 1B), well captured by a logistic fit as $TE_{M\to x(A)} = T_{max}/(1 + exp(-\alpha(A - A_0)))$, with R=0.983. This plateau reflects the natural ceiling set by the extent to which a unit's dynamics can be explained by a shared environmental signal. At low amplitudes, idiosyncratic fluctuations from differences in the intrinsic growth rates (r_i) dominate; therefore, the mean field has little predictive power. As the amplitude increases, the common modulation

progressively shapes all trajectories, making the mean a stronger predictor of individual behavior. However, once the environmental signal overwhelms individual variability, no further information can be gained, and the predictive relationship has reached the limit imposed by the variance in the environmental drive itself. This is consistent with the observation that, in the high-amplitude regime, the fraction of variance in individual trajectories explained by the mean field approaches unity (R^2 ~1; Figure 2A). In other words, stronger modulation eventually adds no new structure and rescales only the same signal. The same saturation trend is reflected in the regression-based measures of alignment (Figure 2B), where both the fraction of variance explained and the signal-to-noise ratio (SNR) – defined as the variance of Mn divided by the variance of the regression residuals – increase with amplitude before leveling off.

Transfer entropy from the mean field to individual units, TE$(M \rightarrow x)$, remains high for slow modulation (long periods) but decreases as the modulation frequency increases (Supplementary Figure S1). This behavior indicates that very rapid environmental oscillations average out relative to the intrinsic chaotic dynamics of the units, preventing the forcing from imposing a coherent population-wide structure. The reverse direction TE$(x \rightarrow M)$ remains near zero across modulation periods.

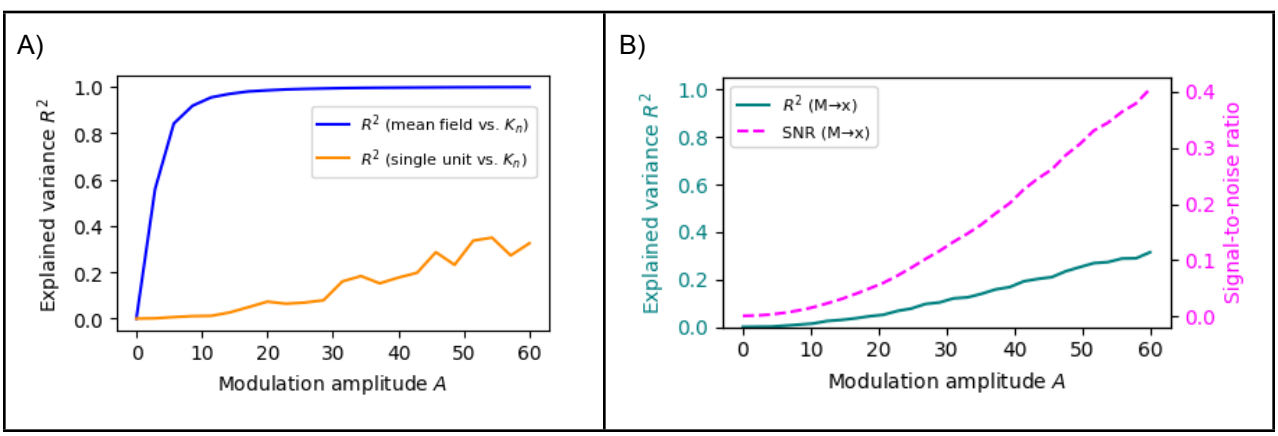

Figure 2. Alignment with mean field and modulation signal under increasing amplitude. (A) Fraction of variance in the mean field (blue) and in a representative unit's trajectory (orange) explained directly by the modulation signal K_n, as a function of A. (B) Relationship between modulation amplitude A and statistical alignment between the mean field M_n and a representative unit $x_{1,n}$. Alignment is quantified using the coefficient of determination (R^2), which represents the fraction of variance in the unit's trajectory

explained by the mean field (teal), and the signal-to-noise ratio (SNR), defined as the variance of M_n divided by the variance of the regression residuals (magenta).

Because the modulation enters the dynamics through the carrying capacity $K_n = K_0 + A\sin(\omega n)$, the effective strength of the forcing is determined by the ratio A/K_0. Repeating the simulations for different baseline carrying capacities (K_0 = 50, 100, 200, 400) shows that the TE$(M \to x)$ curves collapse when plotted as a function of A/K_0 (Supplementary Figure S2), indicating that the relative modulation amplitude acts as the relevant control parameter governing the growth of transfer entropy. The reverse, TE$(x \to M)$, remains near zero across A/K_0 values. The widening of the confidence interval (Figure 1B) at large amplitudes ($A \gtrsim 40 - 50$) reflects increased variability across simulation as the modulation amplitude approaches the scale of the baseline carrying capacity ($A/K_0 \sim 0.5$). In this regime, the strong environmental forcing interacts nonlinearly with the heterogeneous chaotic dynamics of the units, making the resulting trajectories more sensitive to differences in initial conditions and parameter realizations.

We obtained the same qualitative asymmetry when replacing the sinusoidal forcing with a logistic one (representing gradual favorable environmental shift) and with an inverse logistic (representing gradual environmental decline) (Supplementary Figure S4), indicating that the effect is robust to the shape of the modulation. The saturation was not due to numerical clipping; although a soft bound of (0, $2K_0$) = (0, 200) was imposed on all unit states at each time step to prevent unphysical negative concentrations and runaway growth. At the highest modulation amplitude tested (A = 60), the lower bound ($x \leq 0$) was approached in fewer than 10% of unit-timestep pairs across all seeds, and the upper bound ($x \geq 200$) was never reached. Clipping, therefore, cannot account for the TE saturation.

Heuristically, the observed TE asymmetry reflects the growing dominance of shared environmental constraints over independent noise. As A increases, the shared K_n introduces a common pattern into all units, strengthening their correlation with M. Unit-specific fluctuations from intrinsic r values remain uncorrelated and thus average out in the mean. From an information-theoretic standpoint, M is a

low-dimensional projection of the global signal, whereas x contains both the signal and idiosyncratic noise. As the signal fraction increases, the predictive information increases in $M \rightarrow x$ whereas the reverse direction remains negligible. The same trend is reflected in the regression-based measures of alignment between the mean field and a representative unit (Figure B).

Conditional transfer entropy reveals macro-micro asymmetry

The unconditional TE asymmetry demonstrated above could, in principle, arise entirely from the shared environmental signal K_n acting as a common driver. To test this possibility, we computed the conditional transfer entropy (CTE) in both directions while conditioning on K_n. Conditional transfer entropy measures the predictive information provided by the source about the target after accounting for the influence of a third variable. Formally,

$$CTE_{Y \rightarrow X|Z} = \sum p\left(x_{n+1}, x_n^{(k)}, y_n^{(k)}, z_n^{(k)}\right) log_2 \frac{p\left(x_{n+1}|x_n^{(k)}, y_n^{(k)}, z_n^{(k)}\right)}{p\left(x_{n+1}|x_n^{(k)}, z_n^{(k)}\right)},$$

Where $x_n^{(k)}$, $y_n^{(k)}$, and $z_n^{(k)}$ denote the histories of the target, source, and conditioning variable, respectively. In our analysis, the conditioning variable *Z* corresponds to the environmental signal K_n.

Conditioning on K_n substantially reduces the magnitude of the transfer entropy in both directions, indicating that a large portion of the unconditional TE reflects the shared environmental forcing. Nevertheless, a nonnegligible residual information flow persists across modulation amplitudes. At low amplitudes, the conditioned bottom-up term CTE($x \rightarrow M|K_n$) slightly exceeds the macro-to-micro term CTE($M \rightarrow x|K_n$), suggesting that once the shared environmental signal is accounted for, the residual predictive relationship between individual trajectories and the mean field is initially weak. As the modulation amplitude increases, however, the relationship between the conditioned directions reverses, with CTE($M \rightarrow x|K_n$) becoming larger than CTE($x \rightarrow M|K_n$). This transition indicates that, under stronger environmental forcing, the collective dynamics encoded in the mean field retain predictive information

about individual trajectories even after the shared driver is accounted for. This transition is marked by the difference $\Delta_{CTE} = CTE(M \to x|K_n) - CTE(x \to M|K_n)$ changing sign at intermediate amplitudes (A~15).

The results from the conditional analysis indicate that the macro-micro information structure observed in the unconditional analysis has two components: (1) a dominant contribution arising from the shared environmental modulation, and (2) a directional asymmetry that emerges from the collective dynamics of the population.

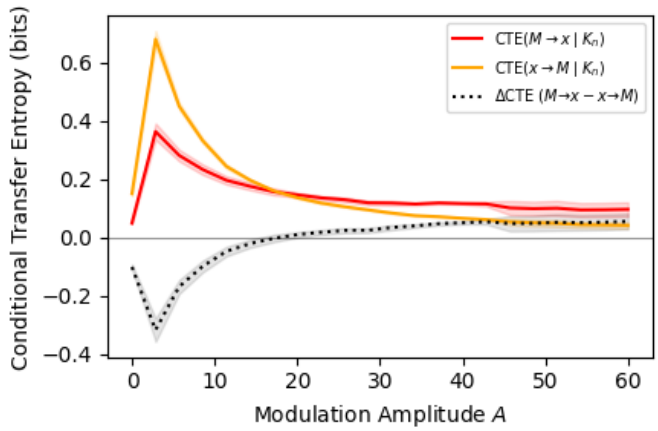

Figure 3. Conditional transfer entropy under shared environmental modulation. Conditional transfer entropy (CTE) between the mean field M_n and a representative unit $x_{i,n}$, conditioning on the environmental signal K_n, shown as a function of modulation amplitude A. CTE$(M \to x|K_n)$ (red) and CTE$(x \to M|K_n)$ (orange) show residual predictive information after accounting for the shared driver. The dotted black curve shows the directional difference $\Delta_{CTE} = CTE(M \to x|K_n) - CTE(x \to M|K_n)$. Results are averaged over 10 seeds (N = 1,000, $K_0 = 100$, $r_i \in [3.9, 4.0]$, and T = 10,000 steps, with the first 1,000 discarded). Shaded regions show 95% confidence intervals.

Inter-unit predictability

The presence of directional information flow between the mean and individuals raises a further question: can shared modulation generate consistent directional relationships between individuals? Even without interactions, a common drive may induce dependencies across the ensemble. We tested this by comparing the transfer entropy (TE) from a reference unit to 50 other units against a null distribution generated from a circularly shifted surrogate series that preserves autocorrelation while destroying temporal alignment (27) (see Methods section). Statistical significance was assessed using an empirical p-value computed from 100 surrogate realizations per target, with a threshold of $p < 0.01$. With no modulation (A=0), no units exceeded the significance threshold, which is consistent with the absence of temporal coordination. Under moderate modulation (A=25), however, 39 out of 50 units were significant, demonstrating that shared modulation alone induces emergent inter-unit predictability.

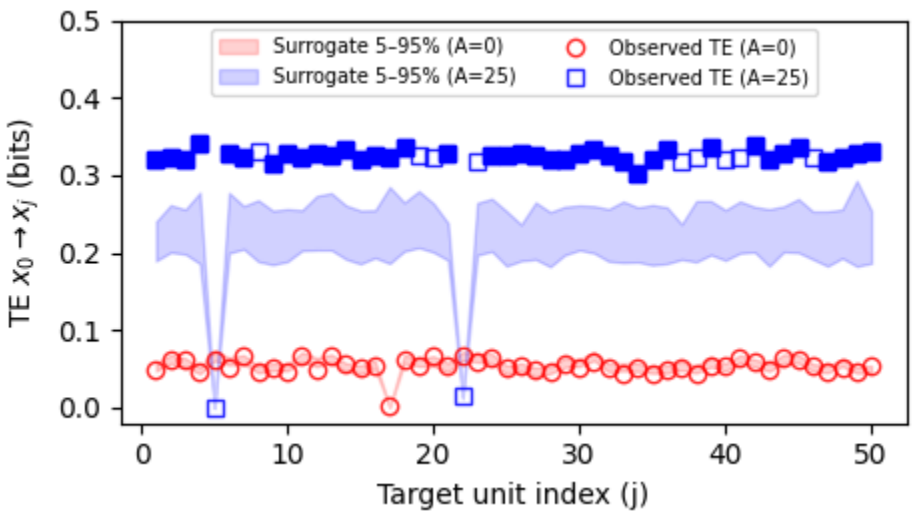

Figure 4. Transfer entropy (TE) from reference unit x_0 to each target unit x_j for two modulation amplitudes ($A = 0$, $A = 25$). Observed TE values (symbols) are compared to the distribution from 100 circularly shifted surrogate source series (shaded bands show the 5-95% percentile range of the surrogate TE distribution). Filled symbols mark significant cases based on the empirical surrogate test ($p < 0.01$); open symbols are non-significant. Results shown for $N = 1000$, $K_0 = 100$, $r_i \in [3.9, 4.0]$, $T = 10,000$ steps (first 1,000 discarded), number of targets analyzed = 50, number of surrogates per target = 100.

Finally, we assessed how modulation shapes macroscopic dynamics by examining the return maps of the mean field. Increasing the modulation amplitude broadens the range of accessible states but maintains trajectories in a coordinated, low-dimensional pattern, indicating that the shared environmental structure expands the dynamical range without disrupting collective organization (Figure S5).

Discussion

Origins-of-life theories often assume that coordination arises from internal chemical coupling among components (e.g., autocatalytic sets (28), hypercycles (29), and early metabolism (30)). These frameworks share the premise that organizations depend on direct chemical interactions; however, they do not always provide satisfactory explanations for the non-trivial question of how such coupling originates in the first place. Here, we explore a simpler scenario in which structured organizations can emerge without direct interactions, driven solely by shared environmental structures.

Our model isolates this idea in its simplest possible form, as a population of uncoupled logistic maps exposed to the same time-varying constraint. Global modulation, implemented through sinusoidal changes in the carrying capacity, introduces a temporal structure without inducing direct interactions between components. We then used transfer entropy (TE) as an operational measure of top-down informational influence to quantify this asymmetry, a framework previously applied to systems with explicit coupling (9). In the transfer entropy (TE) literature, correlations induced by a shared input are often classified as “common driver effects” (20,22,25) and treated as confounds to be removed (23,26). Here, however, shared input is the mechanism of interest. In this context, TE is not intended to imply mechanistic causation, but to test whether shared environmental modulation alone can induce directional predictability between scales. In other words, we ask whether a ‘big picture’ view of the system can forecast what the parts will do next, simply because they are all nudged in the same way. Despite the absence of coupling, the system develops directional asymmetry, and the behavior of the macroscopic mean becomes increasingly predictive of individual dynamics, whereas the reverse direction remains negligible.

The conditional transfer entropy (CTE) analysis clarifies the source of this asymmetry. Conditioning on the environmental signal reduces the magnitude of the transfer entropy, indicating that a substantial portion of the observed TE reflects the shared environmental forcing. Crucially, however, the directional asymmetry does not disappear. The conditioned macro-to-micro term, CTE($M \rightarrow x|K_n$), remains non-zero across amplitudes and becomes larger than the reverse direction, CTE($x \rightarrow M|K_n$), at moderate and high modulation strengths. This residual asymmetry indicates that the mean field retains predictive information about individual dynamics that cannot be attributed solely to the shared environmental signal. Our results therefore suggest that the TE asymmetry reflects two complementary mechanisms: environmentally scaffolded coordination, which dominates under strong forcing, and a weaker emergent component arising from nonlinear aggregation of the population dynamics.

It is important to distinguish two modes of coordination that are often conflated under the umbrella of "top-down causation". Externally imposed boundary conditions - such as the modulated carrying capacity K_n in our model - constrain the dynamics of all units from outside the system. By contrast, internally generated coordination arises from coupling among the units themselves. Our model demonstrates that external boundary conditions alone can produce the informational signatures of top-down influence, including a directional TE asymmetry and inter-unit predictability. The CTE analysis refines this interpretation by separating the component attributable to the external driver from the smaller residual component generated by nonlinear aggregation within the population. This distinction clarifies that the "top-down" signal in our model is not a claim of autonomous macroscopic agency, but rather a demonstration that externally structured environments can generate the informational preconditions from which internally coordinated systems may subsequently emerge.

Although the model is minimal and abstract in nature, it directly relates to scenarios in origins-of-life research where prebiotic chemistry is unfolded under rhythmic environmental forcing. Regular cycles such as wet-dry, freeze-thaw, day-night, and tidal rhythms are known to concentrate reactants, drive condensation, promote molecular restructuring, stabilize intermediates, and facilitate protocell assembly (10-12,31-35). Beyond enabling specific reactions, such cycles may impose system-wide constraints, thus

aligning otherwise independent chemical processes. We represent these well-studied periodic influences with sinusoidal forcing, however, the results are not limited to perfect periodicity. Logistic and inverse-logistic modulation, which represent gradual environmental improvements or declines, yield comparable qualitative asymmetries. The key factor is not the waveform itself, but its capacity to impose a coherent structure across the system. In our framework, modulated carrying capacity serves as a proxy for this type of global context, acting as a constraint capable of entraining independent components without direct, physical interaction. In this sense, the environmental component of TE represents coordination imposed by the shared external context, while the residual component reflects additional predictive structure emerging from the collective dynamics themselves.

This broader principle may apply across many of life's major evolutionary transitions, where collections of simpler units become integrated wholes (in part) due to a shared environment. Structured environmental conditions could have served as an early coordinating force prior to the evolution of internal control mechanisms. For example, vesicle populations may have synchronized division under thermal cycles, early microbial communities may have differentiated along spatial gradients, and hominins may have aligned behaviors through shared cues before developing language. While these examples are speculative, they illustrate how directional information flow can be externally imposed and later internalized by evolving systems. More generally, the findings highlight how shared external drivers can generate proto-organization in uncoupled systems, a mechanism that may recur across the physical, chemical, and biological domains. Our results suggest that such environmentally driven coordination, even if transient, may provide a scaffold for the emergence of persistent self-organized complexity. Future work could test these principles in models incorporating more realistic prebiotic chemistry or biophysics—for example, protocell populations with explicit metabolic kinetics evolving under cyclic environmental forcing—and evaluate whether macro-micro information asymmetries arise when information flow is measured between collective chemical states and individual protocell dynamics.

Methods

Information-theoretic measures

To assess whether shared modulation produces directional relationships between macro- and micro-level dynamics, we computed the transfer entropy (TE) in both directions: top-down (from mean field to representative unit, $M \rightarrow x$) and bottom-up (from representative unit to mean field, $x \rightarrow M$). Formally:

$$T_{Y \rightarrow X} = \sum p\left(x_{n+1}, x_n^{(k)}, y_n^{(k)}\right) log_2 \frac{p\left(x_{n+1} | x_n^{(k)}, y_n^{(k)}\right)}{p\left(x_{n+1} | x_n^{(k)}\right)}$$

where $x_n^{(k)}$ and $y_n^{(k)}$ are the histories of length *k* for the target and source variables, respectively.

To disentangle environmental forcing from genuine mean-field-driven information flow, we computed the conditional transfer entropy (CTE) in both directions, conditioning on the environmental signal K_n:

$$CTE_{Y \rightarrow X|Z} = \sum p\left(x_{n+1}, x_n^{(k)}, y_n^{(k)}, z_n^{(k)}\right) log_2 \frac{p\left(x_{n+1} | x_n^{(k)}, y_n^{(k)}, z_n^{(k)}\right)}{p\left(x_{n+1} | x_n^{(k)}, z_n^{(k)}\right)}$$

where $z_n^{(k)}$ is the history of the conditioning variable K_n. CTE measures the information that the source provides about the target beyond what is already explained by both the target's own past and the conditioning variable's past. If the TE asymmetry is entirely due to the common environmental driver, CTE should be approximately zero in both directions.

Regression-based alignment metrics

We quantified the statistical alignment between the mean field M_n and a representative unit $x_{i,n}$ using two regression-based metrics. First, the coefficient of determination R^2 was obtained from a linear regression of the unit trajectory on the mean field: $\hat{x}_{i,n} = \beta_0 + \beta_1 M_n$. The coefficient of determination (R^2) was computed as:

$$R^2 = 1 - \frac{\sum_n (x_{i,n} - \hat{x}_{i,n})^2}{\sum_n (x_{i,n} - \bar{x}_i)^2}$$

where $\hat{x}_{i,n}$ is the regression prediction and $\bar{x}_i$ is the mean of the unit's trajectory $x_{i,n}$.

Second, the signal-to-noise ratio (SNR):

$$SNR = Var(M_n)/Var(x_{i,n} - \hat{x}_{i,n}),$$

defined as the ratio of the variance of the mean field to the variance of the regression residuals. Both metrics were computed for each amplitude A using ordinary least-squares regression.

Model parameters

The growth rate interval $r_i \in [3.9, 4.0]$ ensures that all units operate deep within the chaotic regime of the logistic map (9,14,16), guaranteeing sensitive dependence on initial conditions and preventing periodic orbits that would trivially synchronize. The baseline carrying capacity was set to $K_0 = 100$. Because the logistic map dynamics depend on the ratio x/K rather than absolute values, results are invariant to this scale. The modulation amplitude A was varied from 0 to 60, spanning the range from no modulation to strong environmental forcing (A/K_0 up to 0.6). The modulation frequency $\omega = 2\pi/2000$ was chosen to provide approximately 4.5 full modulation cycles within the analysis window (robustness to ω is demonstrated in Supplementary Figure S1). The population size $N = 1000$ ensures that the mean field represents a well-averaged macroscopic quantity ($\sqrt{N}$ averaging reduces idiosyncratic noise by a factor of ~31.6). Each simulation was run for $T = 10,000$ time steps, with the first 1,000 steps discarded as transients, leaving $T_{eff} = 9000$ time points for analysis.

Analysis parameters

All information-theoretic quantities (TE and CTE) were computed using a histogram-based (plug-in) method. Each time series was discretized into 15 equally spaced bins, with bin edges determined from the pooled range of the past and future target values. The embedding lag was set to $k = 2$ (rather than the value producing the maximum TE at $k = 1$) to reduce near-synchronous correlations that can inflate TE values (Supplementary Figure S3 shows the sensitivity of TE to the choice

of *k*). Transfer entropy values were expressed in bits by dividing the natural-log mutual information estimates by *ln(2)*. Results were averaged over 10 independent random seeds, each generating a fresh population of growth rates r_i ~ Uniform[3.9, 4.0]. Confidence intervals (95%) were computed using the Student's t-distribution.

Surrogate significance test

To assess whether observed inter-unit transfer entropy values exceeded those expected from temporally misaligned processes, we employed a circular-shift surrogate method (27). For each source-target pair, the source time series was circularly shifted by a random offset (minimum shift of 10 time steps) to preserve its autocorrelation structure while destroying temporal alignment with the target. For each pair, 100 surrogate TE values were generated. Statistical significance was evaluated using an empirical p-value, $p = (1 + n_{sur}(TE_{sur} \geq TE_{obs}))/(1 + N_{sur})$, where TE_{obs} is the TE computed from the original time series, TE_{sur} denotes the surrogate values, $n_{sur}(TE_{sur} \geq TE_{obs})$ is the number of surrogate TE values greater than or equal to the observed value, and $N_{sur} = 100$ is the total number of surrogates. A significance threshold of $p < 0.01$ was used.

Data availability

Code used in the modeling and subsequent analysis is available at https://github.com/celiablanco/macro_to_micro

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